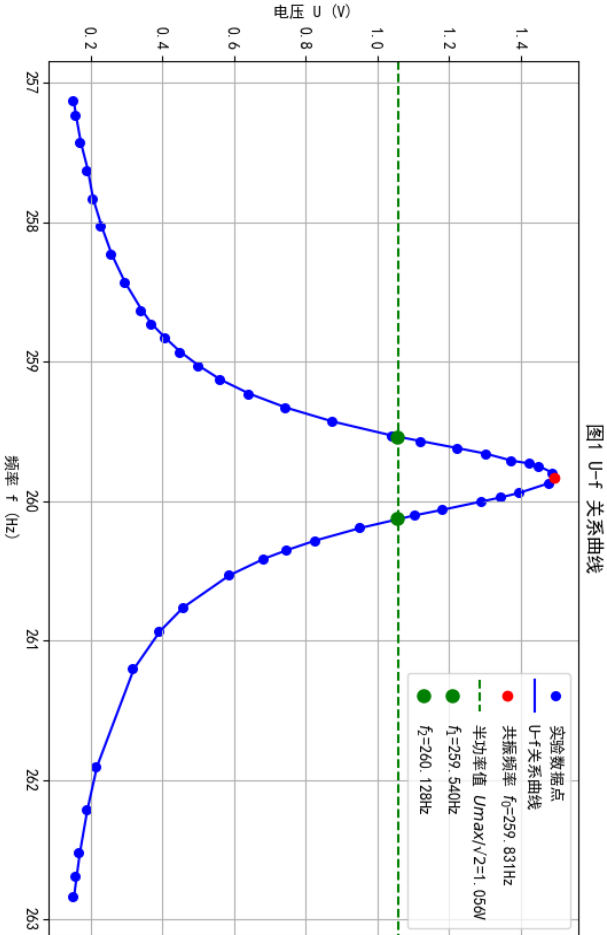
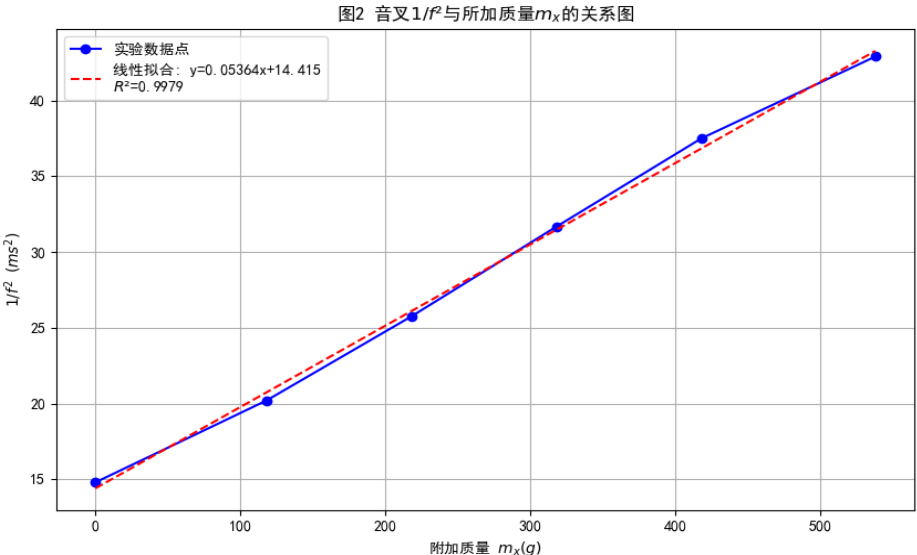


| 表1 音叉的速度-幅频特性实验数据表格 |         |         |         |         |         |         |         |         |  |  |  |
|---------------------|---------|---------|---------|---------|---------|---------|---------|---------|--|--|--|
| f (Hz)              | 257.131 | 257.231 | 257.431 | 257.631 | 257.831 | 258.031 | 258.231 | 258.431 |  |  |  |
| U (V)               | 0.149   | 0.155   | 0.169   | 0.189   | 0.203   | 0.227   | 0.255   | 0.292   |  |  |  |
| f (Hz)              | 258.631 | 258.731 | 258.831 | 258.931 | 259.031 | 259.131 | 259.231 | 259.331 |  |  |  |
| U (V)               | 0.339   | 0.367   | 0.405   | 0.447   | 0.499   | 0.56    | 0.64    | 0.743   |  |  |  |
| f (Hz)              | 259.431 | 259.531 | 259.571 | 259.621 | 259.661 | 259.711 | 259.731 | 259.751 |  |  |  |
| U (V)               | 0.874   | 1.039   | 1.12    | 1.222   | 1.301   | 1.373   | 1.423   | 1.449   |  |  |  |
| f (Hz)              | 259.801 | 259.831 | 259.871 | 259.94  | 259.972 | 260.002 | 260.062 | 260.102 |  |  |  |
| U (V)               | 1.489   | 1.494   | 1.477   | 1.395   | 1.345   | 1.29    | 1.18    | 1.102   |  |  |  |
| f (Hz)              | 260.192 | 260.282 | 260.352 | 260.412 | 260.532 | 260.762 | 260.933 | 261.203 |  |  |  |
| U (V)               | 0.951   | 0.825   | 0.744   | 0.682   | 0.584   | 0.455   | 0.39    | 0.317   |  |  |  |
| f (Hz)              | 261.903 | 262.214 | 262.524 | 262.694 | 262.834 |         |         |         |  |  |  |
| U (V)               | 0.214   | 0.187   | 0.166   | 0.157   | 0.15    |         |         |         |  |  |  |



| 表3 波腹数与驻波频率的关系<br>(张力T=2N, 弦长L=30cm, 弦线直径Φ=0.25mm) |       |        |
|----------------------------------------------------|-------|--------|
| 波腹数 (个)                                            | 波长 cm | 频率 Hz  |
| 1                                                  | 60    | 124.00 |
| 2                                                  | 30    | 226.70 |
| 3                                                  | 20    | 346.50 |

| 表2 音叉的共振频率f与所加质量mx之间关系 |         |         |         |         |         |         |
|------------------------|---------|---------|---------|---------|---------|---------|
| mx(g)                  | 0       | 5g x 2  | 10g x 2 | 15g x 2 | 20g x 2 | 25g x 2 |
| f(Hz)                  | 259.834 | 222.431 | 197.038 | 177.668 | 163.299 | 152.659 |
| U(v)                   | 1.490   | 1.315   | 1.253   | 1.123   | 0.995   | 0.112   |
| 1/f²(ms²)              | 14.810  | 20.210  | 25.760  | 31.679  | 37.500  | 42.900  |



| 表4 传播速度与张力的关系<br>(弦长L=30cm, 弦线直径Φ=0.25mm) |      |         |         |       |          |
|-------------------------------------------|------|---------|---------|-------|----------|
| 砝码质量 g                                    | 张力 N | 波腹数 (个) | 驻波频率 Hz | 波长 cm | 传播速度 m/s |
| 150                                       | 1.5  | 1       | 111.50  | 60    | 66.90    |
| 200                                       | 2.0  | 1       | 124.30  | 60    | 74.58    |
| 250                                       | 2.5  | 1       | 137.70  | 60    | 82.62    |
| 300                                       | 3.0  | 1       | 148.00  | 60    | 88.80    |